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Enclosure with locally-flexible regions

Abstract

A force input/haptic output interface for an electronic device can include a force input sensor and a haptic actuator. In one example, the force input sensor and the haptic actuator are accommodated on a frame positioned below an input surface. In many examples, the frame includes relieved portions that redirect and/or concentrate compression or tension in the haptic actuator into the frame.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This is a continuation of U.S. patent application Ser. No. 16/989,657, filed 10 Aug. 2020, and entitled "ENCLOSURE WITH LOCALLY-FLEXIBLE REGIONS," which is a continuation of U.S. patent application Ser. No. 15/657,040, filed 21 Jul. 2017, entitled "ENCLOSURE WITH LOCALLY-FLEXIBLE REGIONS," now U.S. Pat. No. 10,775,889, issued 15 Sep. 2020, the disclosures of which are hereby incorporated by reference in their entireties.

FIELD

(1) Embodiments described herein relate to user interfaces for electronic devices, and in particular, to electronic device enclosures that include a distribution of locally-flexible regions that can be coupled to haptic actuators to provide haptic output to that user.

BACKGROUND

(2) An input sensor for an electronic device can detect when a user applies a purposeful force, generally referred to as a "force input," to a surface of the electronic device. Such sensors, together with associated circuitry and structure, can be referred to as "force input sensors."

(3) A mechanical actuator for an electronic device can generate a mechanical output, generally referred to as a "haptic output," through a surface of the electronic device. Such actuators, together with associated circuitry and structure, can be referred to as "haptic actuators."

(4) In some cases, an electronic device can associate a force input sensor and a haptic actuator with the same surface, generally referred to as an "user interface surface." Conventionally, a user interface surface, such as a trackpad of a laptop computer, extends through an opening defined in an enclosure of the electronic device. However, as a result of the opening, the enclosure of the electronic device may be undesirably structurally weakened, increasing manufacturing complexity and susceptibility of the electronic device to damage.

SUMMARY

(5) Embodiments described generally reference an electronic device that includes a force input/haptic output interface integrated into, or associated with, an enclosure of the electronic device. More specifically, an electronic device such as described herein is positioned in an enclosure that has an external surface (which may be contiguous) and an interior surface opposite the external surface. In many embodiments, a frame, internal to the enclosure, is coupled to the interior surface. The frame includes a locally-flexible region that is defined, at least in part, by a reduced-thickness section and a support structure adjacent to reduced-thickness section. A force transducer (such as a piezoelectric element) is coupled to the support structure. As a result of this construction, an actuation of the force transducer induces a bending moment into the support structure to generate a haptic output through the external surface.

(6) In some embodiments, more than one locally-flexible region and, correspondingly, more than one force transducer may be associated with the frame. In other embodiments, the frame may be integrated with the interior surface of the enclosure. In many examples, the enclosure is formed from glass, but this is not required.

(7) Further embodiments described generally reference an electronic device including an enclosure formed from glass that accommodates a keyboard in an upper region of an external surface. The enclosure also accommodates a force input/haptic output interface (such as described herein) in a lower region of the external surface, generally below the keyboard. In these examples, locally-flexible regions may not be required; a haptic actuator or a piezoelectric element can be coupled directly to an interior surface of the lower portion of the external surface of the enclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Reference will now be made to representative embodiments illustrated in the accompanying figures. It should be understood that the following descriptions are not intended to limit this disclosure to one preferred embodiment. To the contrary, the disclosure provided herein is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the described embodiments, and as defined by the appended claims.
(2) FIG. **1** depicts an electronic device that can incorporate a force input/haptic output interface, such as described herein.
(3) FIG. **2**A depicts an example distribution of locally-flexible regions defined into an interior surface of an electronic device, such as the electronic device of FIG. **1**, that may be associated with a force input/haptic output interface.
(4) FIG. **2**B depicts a locally-flexible region of the enclosure depicted in FIG. **2**A.
(5) FIG. **2**C depicts another view of the locally-flexible region depicted in FIG. **2**B.
(6) FIG. **2**D depicts a locally-flexible region of the enclosure depicted in FIG. **2**A, taken through line A-A.
(7) FIG. **2**E depicts the locally-flexible region of the enclosure depicted in FIG. **2**D, showing flexion of the locally-flexible region.
(8) FIG. **2**F depicts another example locally-flexible region of an enclosure, such as depicted in FIG. **2**A.
(9) FIG. **2**G depicts the locally-flexible region of the enclosure depicted in FIG. **2**F, showing flexion of the locally-flexible region.
(10) FIG. **3** depicts another example distribution of locally-flexible regions associated with a force input/haptic output interface.
(11) FIG. **4** depicts another example distribution of locally-flexible regions associated with a force input/haptic output interface.
(12) FIG. **5** depicts another example distribution of locally-flexible regions associated with a force input/haptic output interface.
(13) FIG. **6**A depicts a cross-section of an example force transducer coupled to a locally-flexible region of an interior surface of an electronic device enclosure, such as described herein.
(14) FIG. **6**B depicts a cross-section of another example force transducer coupled to a locally-flexible region of an interior surface of an electronic device enclosure, such as described herein.
(15) FIG. **6**C depicts a cross-section of another example force transducer coupled to a locally-flexible region of an interior surface of an electronic device enclosure, such as described herein.
(16) FIG. **7** depicts another example distribution of locally-flexible regions associated with a force input/haptic output interface, such as described herein.
(17) FIG. **8** depicts another example distribution of locally-flexible regions associated with a force input/haptic output interface, such as described herein.
(18) FIG. **9** depicts a cross-section of an example force transducer coupled to a locally-flexible region of an interior surface of an electronic device enclosure, such as described herein.
(19) FIG. **10** depicts a cross-section of another example force transducer coupled to a locally-flexible region of an interior surface of an electronic device enclosure, such as described herein.
(20) FIG. **11** depicts a cross-section of another example force transducer and locally-flexible region of an interior surface of an electronic device enclosure, such as described herein.
(21) FIG. **12** depicts a cross-section of another example force transducer and locally-flexible region of an interior surface of an electronic device enclosure, such as described herein.
(22) FIG. **13** depicts a cross-section of another example force transducer and locally-flexible region of an interior surface of an electronic device enclosure, such as described herein.
(23) FIG. **14** depicts a cross-section of another example force transducer and locally-flexible region of an interior surface of an electronic device enclosure, such as described herein.

(24) FIG. **15** depicts a cross-section of another example force transducer and locally-flexible region of an interior surface of an electronic device enclosure, such as described herein.
(25) FIG. **16** is a flow chart depicting example operations of a method of forming a haptic actuator, such as described herein.
(26) FIG. **17** is a flow chart depicting example operations of a method of providing haptic feedback.
(27) FIG. **18** is a flow chart depicting example operations of a method of receiving force input.
(28) The use of the same or similar reference numerals in different figures indicates similar, related, or identical items.
(29) The use of cross-hatching or shading in the accompanying figures is generally provided to clarify the boundaries between adjacent elements and also to facilitate legibility of the figures. Accordingly, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, element proportions, element dimensions, commonalities of similarly illustrated elements, or any other characteristic, attribute, or property for any element illustrated in the accompanying figures.
(30) Additionally, it should be understood that the proportions and dimensions (either relative or absolute) of the various features and elements (and collections and groupings thereof) and the boundaries, separations, and positional relationships presented therebetween, are provided in the accompanying figures merely to facilitate an understanding of the various embodiments described herein and, accordingly, may not necessarily be presented or illustrated to scale, and are not intended to indicate any preference or requirement for an illustrated embodiment to the exclusion of embodiments described with reference thereto.

DETAILED DESCRIPTION

(31) Embodiments described herein reference an electronic device that includes a force input/haptic output interface. The phrase "force input/haptic output interface," as used herein, generally references a system or set of components configured to receive force input at a surface from a user and, additionally, to provide haptic output to that same user through the same surface. The surface associated with a force input/haptic output interface (such as described herein) can be referred to as an "user interface surface."
(32) In one example, a force input/haptic output interface is operated in conjunction with, and/or positioned over, a display of an electronic device. In this example, the user interface surface can be a protective outer cover (e.g., transparent glass, sapphire, plastic) positioned over an active display area of the display. A user can exert a force onto the protective outer cover to interact with content shown on the display at that location. In response, the force input/haptic output interface can generate a haptic output (e.g., click, vibration, shift, drop, pop, and so on) through or on the display at, or near, that location to inform the user that the force input was received. In other examples, one or more haptic outputs can be provided through the protective outer cover in response to, or independent of, one or more force inputs in a different or implementation-specific or configuration-specific manner.
(33) In another example, a force input/haptic output interface is operated in conjunction with a user interface surface of an electronic device, such as a trackpad. In this example, the user interface surface is a continuous and planar external surface of the trackpad, which can be formed from an opaque or transparent material such as metal, glass, organic materials, synthetic materials, woven materials, and so on. A user can exert a force onto a portion of the user interface surface to instruct the electronic device to perform an action. In response, the force input/haptic output interface can generate a haptic output at, or near, that location to inform the user that the force input was received. As with other example configurations, one or more haptic outputs can be provided through the user interface surface in response to, or independent of, one or more force inputs in a different or implementation-specific or configuration-specific manner.
(34) For simplicity of description, many embodiments that follow reference a force input/haptic

output interface operated in conjunction with a non-display region of a portable electronic device, such as a trackpad region of a laptop computer. In these examples, the user interface surface is a contiguous external surface of the portable electronic device, although this may not be required. It may be appreciated, however, that this is merely one example; other configurations, implementations, and constructions are contemplated in view of the various principles and methods of operation, and alternatives thereto, described in reference to the embodiments that follow.

(35) A force input/haptic output interface (such as described herein) can be implemented with one or more force input sensors and/or one or more haptic actuators. In some cases, a single component, referred to as a “force transducer,” can be configured to provide haptic output and to receive force input. For simplicity of description, certain embodiments that follow reference a force transducer, but it may be appreciated that this is merely one example construction; other embodiments may include separate force input sensors and haptic actuators.

(36) A force input/haptic output interface can be implemented with a set of force transducers coupled to an interior surface of an enclosure of an electronic device. Actuation of a force transducer induces a haptic output through an exterior surface of the enclosure, opposite the interior surface. Similarly, a force applied to the exterior surface of the enclosure can locally deform the interior surface. In response to the local deformation, the force transducer can generate or change a signal in a manner corresponding to the local deformation. The signal, in turn, can be correlated to a force input (e.g., a magnitude, direction, and/or location of force applied to the exterior surface).

(37) It may be appreciated that the thickness of the enclosure separating the interior surface from the exterior surface can affect one or more characteristics of a haptic output generated and/or one or more characteristics of a force input received. More specifically, the thicker the enclosure, the more attenuated haptic outputs and force inputs may be.

(38) In many embodiments, an enclosure of an electronic device can be formed to a structural thickness sufficient to support, enclose, and/or contain components and elements of an electronic device. An interior surface of the enclosure can be defined by regions that are thinned, stiffened, or supported in a manner that confers specific mechanical properties to those regions of the interior surface, such as greater local flexibility or greater local stiffness.

(39) In one example, an interior surface of an enclosure includes multiple locally-flexible regions. A locally-flexible region can include one or more cavities, openings, perforations, or reduced-thickness sections that at least partially surround (or circumscribe) and define a support structure (e.g., a fixed-fixed beam having two ends, each of which are constrained). A haptic actuator, as an example of a force transducer, can be coupled to the support structure such that compression or expansion of the haptic actuator (parallel or perpendicular to the interior surface) induces a bending moment (e.g., a deformation) in the support structure which, in turn, induces a haptic output through or on the external surface of the electronic device enclosure. The degree to which the support structure bends in response to actuation of the haptic actuator may be defined or controlled, at least in part, by the geometry of the reduced-thickness sections.

(40) For example, the thinner a reduced-thickness section is, the more the support structure within the locally-flexible region may bend or otherwise deform. In some cases, a reduced-thickness section can include one or more openings or perforations, but this may not be required of all embodiments. In many embodiments, the reduced-thickness sections have a thickness that is less than the support structure and less than the enclosure. In some examples, the support structure may have a thickness that is less than a thickness of the enclosure, but this may not be required of all embodiments. In still further embodiments, locally-flexible regions may not be required and a force transducer may be coupled directly to the interior surface of the electronic device enclosure.

(41) In still further embodiments, the enclosure may have a substantially constant thickness. In these examples, the enclosure may be locally strengthened by a frame coupled to the interior surface of the enclosure. In this manner, regions of the interior surface of the enclosure that are not coupled to the frame may be more flexible (e.g., locally-flexible) than regions that are supported by

the frame.
(42) These and other embodiments are discussed below with reference to FIGS. **1-18**. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanation only and should not be construed as limiting.
(43) FIG. **1** shows an electronic device **100** that can include a force input/haptic output interface, such as described herein. As with other embodiments, the force input/haptic output interface can be configured to receive force input from a user and to provide haptic output to that same user. In some examples, the force input/haptic output interface is associated with a display of the electronic device **100**. For example, the force input/haptic output interface may be positioned behind or along a perimeter of the display. In other examples, the force input/haptic output interface is associated with an input area of an enclosure the electronic device **100**, such as a trackpad area adjacent to a keyboard area of an enclosure of a laptop computer.
(44) For simplicity of description and illustration, the electronic device **100** is depicted in FIG. **1** as a laptop computer having a force input/haptic output interface integrated into a trackpad. However, it may be appreciated that this is merely one example and that other implementations of force input/haptic output interfaces can be integrated into, associated with, or take the form of different components or systems of other electronic devices including, but not limited to: desktop computers; tablet computers; cellular phones; wearable devices; peripheral devices; input devices; accessory devices; cover or case devices; industrial or residential control or automation devices; automotive or aeronautical control or automation devices; a home or building appliance; a craft or vehicle entertainment; control; and/or information system; a navigation device; and so on.
(45) In the illustrated example, the electronic device **100** includes an enclosure **102** to retain, support, and/or enclose various electrical, mechanical, and structural components of the electronic device **100**, including a primary display **104**, a keyboard **106**, and a secondary display **108**. The enclosure **102** can be formed from, as an example, glass, sapphire, ceramic, metal, or plastic, or any combinations thereof. The electronic device **100** can also include a processor, memory, power supply and/or battery, network connections, sensors, input/output ports, acoustic elements, haptic actuators, digital and/or analog circuits for performing and/or coordinating tasks of the electronic device **100**, and so on. For simplicity of illustration, the electronic device **100** is depicted in FIG. **1** without many of these elements, each of which may be included, partially and/or entirely, within the enclosure **102** and may be operationally, structurally, or functionally associated with, or coupled to, the primary display **104**, the keyboard **106**, the secondary display **108**, and/or a force input/haptic output interface **110**.
(46) The force input/haptic output interface **110** includes a set of force transducers distributed relative to a user interface surface that may be touched by a user. In the illustrated embodiment, the user interface surface is positioned in, or projects from, a rectangular opening defined through a lower portion of the enclosure **102**. In other embodiments, the user interface surface can extend across the width of the lower hinged portion of the enclosure **102**. In still further embodiments, the opening and/or user interface surface can take another shape.
(47) In the illustrated example, the user interface surface associated with the force input/haptic output interface **110** is separate from the enclosure **102**. More particularly, the user interface surface is positioned in an opening defined through the enclosure **102**. The opening is depicted as a rounded rectangle, but may take any suitable shape.
(48) The user interface surface can be formed from any number of suitable materials. In some examples, the user interface surface is formed from the same material (or a similar material) as the enclosure **102**, but this may not be required. For example, in one embodiment, the enclosure **102** is formed from metal and the user interface surface is formed from glass. In another example, the enclosure **102** is formed from glass and the user interface surface is formed from a ceramic material. In other examples, the user interface surface may be integrated into the enclosure **102**. In many examples, the user interface surface is associated with another interface or input system of

the electronic device **100**, such as a touch input system.
(49) In some cases, the user interface surface can be integrated into or otherwise be a part of the enclosure **102**. In other words, an opening defined through the enclosure **102**, such as shown, does not exist; the set of force transducers associated with the force input/haptic output interface **110** is coupled directly to an interior surface of the enclosure **102**.
(50) Each force transducer of the set of force transducers associated with the force input/haptic output interface **110** can be arranged relative to the user interface surface in a number of ways. For example, in some embodiments, the set of force transducers is arranged in a grid and includes four separate force transducers. In other cases, the set of force transducers contains five separate force transducers arranged in multiple, offset, rows. In some cases, each force transducer has the same shape, whereas in others certain force transducers may be larger or smaller than others. More or fewer force transducers may be use in any configuration and/or embodiment described herein.
(51) Similarly, each force transducer associated with the force input/haptic output interface **110** can be coupled to the user interface surface in any of a number of ways. For example, in some embodiments, each force transducer is coupled directly to the user interface surface using an adhesive. In other cases, each force transducer is coupled to a substrate (e.g., a glass sheet) or frame that, in turn, is coupled to the user interface surface.
(52) Generally and broadly, FIGS. **2**A-**6**C depict various example constructions of a force input/haptic output interface. More specifically, these embodiments generally take the form of a force input/haptic output interface that includes haptic actuators that are coupled to locally-flexible regions defined into an interior surface of an electronic device enclosure. In these embodiments, the locally-flexible regions are each defined, at least part, by a reduced-thickness section that in turn defines a complete or partial perimeter of a support structure that is coupled to a haptic actuator. In this manner, actuation of the haptic actuator induces a bending moment and/or other deformation in the support structure.
(53) In one example, a locally-flexible region includes two parallel rectilinear reduced-thickness sections defining a support structure between them. In this manner, the support structure takes the form of a bending beam that is fixed on two ends. In this example, the support structure has a thickness greater than that of the reduced-thickness sections, but this may not be required of all embodiments.
(54) In another example, a locally-flexible region includes two parallel rectilinear openings or apertures defining a support structure between them. In other words, the two rectilinear openings are aligned with each other and offset from each so as to define a support structure having particular flexibility or rigidity. In this manner, the support structure takes the form of a bending beam that is fixed on two ends.
(55) In another example, a locally-flexible region includes one reduced-thickness section that defines three edges of a rectilinear a support structure. In this manner, the support structure takes the form of a cantilevered beam that is fixed on one end.
(56) In yet another example, a locally-flexible region includes four reduced-thickness sections arranged in a grid, defining a cross-shaped support structure between them. In yet another example, a locally-flexible region includes curved reduced-thickness sections. In yet another example, a locally-flexible region includes a number of perforations in place of a reduced-thickness section. In yet another example, a locally-flexible region includes a reduced-thickness section that entirely circumscribes a support structure. In this manner, the support structure takes the form of an island surrounded entirely by a reduced-thickness section.
(57) Accordingly, generally and broadly, a locally-flexible region such as described herein typically includes at least one reduced-thickness section (that may be or include an aperture) that defines at least a portion of a perimeter of a beam or support structure. A haptic actuator is typically coupled to the beam or support structure.
(58) For simplicity of description and illustration, the embodiments that follow reference one

example construction of a locally-flexible region including two substantially parallel rectilinear reduced-thickness sections defining a support structure between them. It may be appreciated, however, that this is not required and a reduced-thickness section (or aperture or opening) and/or a locally-flexible region can be suitable configured differently in different embodiments.

(59) For example, FIG. **2**A depicts an example distribution of locally-flexible regions formed or defined into an interior surface of an enclosure. The local-flexible regions are shown in dashed lines as these regions are not normally visible in the view depicted in FIG. **2**A. The locally-flexible regions can be associated with a force input/haptic output interface incorporated into an electronic device **200**. In particular, the electronic device **200** includes an enclosure **202** that defines an external surface **204**. The external surface **204** may be associated with a respective interior surface of the enclosure **202**. The external surface **204** may be contiguous and planar, although this is not required.

(60) In the illustrated embodiment, the external surface **204** is shown with a region that generally extends parallel to a length of a keyboard, such as shown in FIG. **1**, although this configuration is not required. In other embodiments, the external surface **204** can be configured in another manner.

(61) For example, in other cases, the external surface **204** can be defined elsewhere, relative to the enclosure **202**. For example, the external surface **204** may be associated with a portion of the enclosure **202** generally above the depicted keyboard (as used herein terms such as "above" and "below" are relative to a typical orientation of an electronic device, such as the electronic device **200**, when in use). In other cases, the external surface **204** may be below the enclosure **202**, on an underside of the electronic device. In still other cases, the external surface **204** may be defined in a sidewall or edge of the enclosure **202**. It may be appreciated that the external surface **204**, associated with the force input/haptic output interface, can be suitable configured in or incorporated into any suitable surface of the enclosure **202**.

(62) In this embodiment, the external surface **204** defines an opening to accommodate a user interface surface **206**, which may be formed from a different material than the external surface **204**. The opening is aligned approximately in the center of the first region (e.g., a lower region) of the external surface **204** and extends approximately half a width of the lower region.

(63) In the illustrated embodiment, four locally-flexible regions **208**, **210**, **212**, **214** are illustrated in phantom and defined into the interior surface of the enclosure **202**. The four locally-flexible regions are distributed in a two-by-two grid.

(64) The four locally-flexible regions are typically configured and constructed in the same manner, but this is not required. For example, in some embodiments, the locally-flexible region **208** and the locally-flexible region **210** are configured as a first pair of locally-flexible regions sharing one or more flexibility or rigidity properties, whereas the locally-flexible region **212** and the locally-flexible region **214** are configured as a second pair of locally-flexible regions sharing one or more flexibility or rigidity properties that are different than the properties of the first pair. For simplicity of description, the description that follows references the locally-flexible region **208**; it is appreciated that the locally-flexible regions **210**, **212**, and **214** may be similarly configured.

(65) Turning to FIGS. **2**B-**2**E, the locally-flexible region **208** defines a support structure **216** to support a haptic actuator **218**, such as a piezoelectric element. The support structure **216** is bordered by two reduced-thickness sections, identified as the reduced-thickness sections **220***a* and **220***b*. The reduced-thickness sections **220***a* and **220***b* have a thickness less than that of the enclosure **202** and less than that of the support structure **216**. The reduced-thickness sections **220***a* and **220***b* can be formed into the interior surface of the enclosure **202** in any suitable manner including, but not limited to: ablation; etching; stamping; scribing; and so on. In some examples, the reduced-thickness sections **220***a* and **220***b* include one or more apertures or perforations (not shown). The interior surface of the enclosure **202** is identified in FIGS. **2**B-**2**C as the interior surface **222**.

(66) In the illustrated example, and as shown in FIGS. **2**B-**2**C, the support structure **216** is a

rectilinear bending beam with two fixed ends, a first end **216***a* and a second end **216***b*. (see, e.g., FIG. **2**A and FIGS. **2**D-**2**E). However, this is merely one example. In other embodiments, the support structure **216** can take any number of suitable shapes including, but not limited to: a cross shape; a circular shape; a curved shape; a spoke-and-hub shape; and so on.
(67) In some cases, the support structure **216** can have a thickness that is less than that of the enclosure **202** and/or the user interface surface **206**, although this may not be required. For example, as illustrated (see FIGS. **2**B-**2**C), the support structure **216** has a thickness less than that of the enclosure **202**.
(68) As a result of this construction, compression or expansion of the haptic actuator **218** in a direction parallel to the user interface surface **206** induces a bending moment, deforming either toward the user interface surface **206** or toward an interior volume within the enclosure **202**, in the support structure **216**. FIG. **2**E depicts an outward deformation of the user interface surface **206** as a result of a compression of the haptic actuator **208218**. More specifically, parallel compression of the haptic actuator **208218** (e.g., parallel to the user interface surface **206**) results in perpendicular deformation of the user interface surface **206**. In other cases, the user interface surface **206** may deform inwardly. In still other embodiments, the user interface surface **206** may deform both outwardly and inwardly (e.g., oscillation or vibration).
(69) In another example, the haptic actuator **218** can compress or expand in a direction perpendicular to the user interface surface **206**. For example, as illustrated in FIG. **2**F, an enclosure **224** can include a locally-flexible region **226** of an external user interface surface. The locally-flexible region **226** can be positioned opposite an internal frame **228**. A haptic actuator **230** is positioned between the internal frame **228** and the locally-flexible region **224**.
(70) As a result of this construction, compression or expansion of the haptic actuator **230** in a direction perpendicular to the locally-flexible region **226** induces a bending moment, deforming the locally-flexible region **226** either outward or, alternatively, toward an interior volume within the enclosure **224**, in the support structure **216**. FIG. **2**G depicts an outward deformation of the locally-flexible region **226** as a result of an expansion of the haptic actuator **230**. More specifically, perpendicular expansion of the haptic actuator **230** (e.g., perpendicular to the locally-flexible region **226**) results in perpendicular deformation of the locally-flexible region **226**. In other cases, the locally-flexible region **226** may deform inwardly. In still other embodiments, the locally-flexible region **226** may deform both outwardly and inwardly (e.g., oscillation or vibration).
(71) Other embodiments can be implemented in another manner. For example, FIG. **3** depicts another example distribution of force transducers, each associated with a respective locally-flexible region defined into an interior surface of the enclosure, and that can be associated with a force input/haptic output interface incorporated into an electronic device **300**. In particular, the electronic device **300** includes an enclosure **302** that defines an external surface **304**. The external surface **304** defines an upper region and a lower region. In this embodiment, the external surface **304** defines the user interface surface; no opening or separate layer to define the interface is required. In this manner, the visual continuity of the external surface **304** is not interrupted. The locally-flexible regions can be formed into an interior surface of the enclosure **302** below the lower region of the external surface **304**.
(72) In the illustrated embodiment, the interior surface of the enclosure **302** defines ten locally-flexible regions associated with ten force transducers, arranged in two parallel rows (e.g., each local-flexible region positioned in a selected location). One of the force transducers is labeled as the force transducer **306**. As with other embodiments described herein, the force transducer **306** is coupled to a locally-flexible region defined into the interior surface of the enclosure **302**. In some embodiments, the locally-flexible region is defined into or defined by a frame coupled to the interior surface of the enclosure **302**.
(73) In the illustrated example, the frame includes reduced-thickness sections that cooperate to define a rectilinear bending beam (identified as the support structure **308**) having two fixed ends,

identified as the fixed ends **308***a* and **308***b*. In the illustrated embodiment, the reduced-thickness sections are identified as the reduced-thickness sections **310***a* and **310***b*. In this manner, the frame defines locally-flexible region similar to the local-flexible regions referenced with respect to other embodiments described herein.

(74) A piezoelectric element **312** is positioned on the support structure **308**. In other embodiments, the support structure **308** can take any number of suitable shapes including, but not limited to: a cross shape, a circular shape, a curved shape, a spoke-and-hub shape, and so on.

(75) As a result of the depicted construction, compression or expansion of the piezoelectric element **318** induces a bending moment within the support structure **308**, either toward the external surface **304** of the enclosure **302** or toward an interior volume within the enclosure **302**.

(76) Still further embodiments can be implemented in another manner. For example, FIG. **4** depicts an electronic device **400** that includes another example distribution of force transducers that can be associated with a force input/haptic output interface. In particular, the electronic device **400** includes an enclosure **402** that defines a user interface surface **404** that extends across an entirety of a width of a lower portion of the enclosure **402**, substantially parallel to a length of a keyboard positioned in an upper portion of the enclosure **402**.

(77) In the illustrated embodiment, the interior surface of the enclosure **402** includes two parallel rows of locally-flexible regions, identified as the upper row **406** and the lower row **408**. In this example, the lower row **408** can include locally-flexible regions that have different mechanical properties than the locally-flexible regions of the upper row **406**. In particular, the locally-flexible regions of the lower row **408** may be more rigid (e.g., smaller in area) than the locally-flexible region of the upper row **406**.

(78) Still other configurations and constructions may be implemented. For example, FIG. **5** depicts an electronic device **500** with another example distribution of force transducers. The electronic device **500** includes an enclosure **502** that defines a user interface surface **504**. In the illustrated embodiment, a set of locally-flexible regions defined into the internal surface of the enclosure **502** supports a number of force transducers, arranged in three parallel rows, one of which is identified as the row **506**.

(79) As noted above with respect to other embodiments, the locally-flexible regions are typically configured and constructed in the same manner, but this is not required. For example, in some embodiments, the row **506** may include a locally-flexible region that exhibit different mechanical properties. For example, a first locally-flexible region **508** may include with a cantilevered **510** (e.g., a cantilevered beam) that is defined by a reduced-thickness section **512**. A second locally-flexible region **514** may be associated with a pair of reduced-thickness sections that cooperate to define a rectilinear bending beam having two fixed ends, such as described with respect to other embodiments herein.

(80) In this example, each row of force transducers includes the same number of force transducers, but configuration this is not required. For example, in some embodiments, different rows of force transducers may include a different number or arrangement of force transducers.

(81) The foregoing embodiments depicted in FIGS. **2**A-**5** and the various alternatives thereof and variations thereto are presented, generally, for purposes of explanation, and to facilitate an understanding of various configurations of a force input/haptic output interface and the various components thereof, such as described herein. However, it will be apparent to one skilled in the art that some of the specific details presented herein may not be required in order to practice a particular described embodiment, or an equivalent thereof.

(82) For example, it may be understood that, generally and broadly, embodiments described herein can arrange any suitable number of force transducers in any number of suitable patterns including, but not limited to: grid patterns; column-and-row patterns; alternating patterns; repeating patterns; tessellated patterns; offset patterns; and so on. Further, it may be appreciated that in certain examples, locally-flexible regions may be included to define or modify particular load paths

through a frame supporting the various force transducers or an interior surface of an electronic device enclosure that supports the various force transducers. In other embodiments, the locally-flexible regions are associated with reduced-thickness sections (e.g., formed by abrasion, ablation, etching, molding, and so on).

(83) Further, it may be understood that, although the foregoing embodiments reference force transducers implemented with piezoelectric elements, this configuration is not required. In other embodiments, other electroactive elements, materials, or constructions may be used. Suitable materials and constructions may include, without limitation: electroactive polymers; shape-change or memory-wire (e.g., nitinol); ferroelectric polymers; microelectromechanical systems; magnetic attractors; linear actuators; solenoid-based systems; and so on.

(84) Further, in many examples, the support structures depicted and described in reference to FIGS. **2**A-**5** may not share a uniform thickness with adjacent portions of the interior surface of an electronic device enclosure. For example, FIGS. **6**A-**6**C generally and broadly depict cross-sections of an example force transducer coupled to a locally-flexible region, such as a frame or an interior surface of an electronic device enclosure.

(85) In particular, FIG. **6**A depicts a cross-section of an example force transducer **600***a* coupled to a locally-flexible region **602** of an enclosure, such as described herein. The cross-section is taken through line A-A of FIG. **2**A, and shows an embodiment different than that of FIGS. **2**B-**2**E. The locally-flexible region **602** includes a relieved section **604** that forms a support structure. The relieved section **604** has a thickness less than that of portions of the locally-flexible region **602** adjacent to the relieved section **604**.

(86) The relieved section **604** can be formed in any number of suitable ways. For example, the relieved section **604** can be formed by laser ablation, mechanical etching, chemical etching, or any combination thereof. A piezoelectric element **606** is coupled to the relieved section **604** such that compression or expansion of the piezoelectric element **606** induces a bending moment in the relieved section **604**. Similarly, bending of the relieved section **604** can result in a concentration of compression or tension in the piezoelectric element **606**.

(87) FIG. **6**B depicts a cross-section of an example force transducer **600***b* coupled to a locally-flexible region **602**, such as described herein. The locally-flexible region **602** includes a first relieved section **610** opposite a second relieved section **612** (e.g., different surface of an enclosure). More specifically, the first relived section **610** may be defined into an external surface of a housing and the second relived section may be defined into an internal surface of the housing. The combination of the first relieved section **610** and the second relieved section **612** results in a thickness less than that of portions of the locally-flexible region **602** adjacent to those sections.

(88) The relieved sections can be formed, and may function, as described with respect to other embodiments described herein (e.g., FIGS. **2**A-**5**).

(89) Although illustrated as opposite one another, one may appreciate that the first relieved section **610** and the second relieved section **612** need not be symmetrically formed. For example, in some embodiments, the second relieved section **612** may be wider than the first relieved section **610**. In other cases, the second relieved section **612** may only partially overlap the first relieved section **610**.

(90) In still further embodiments, a locally-flexible region can be formed by stiffening or supporting one substrate layer (e.g., an outer layer of an electronic device enclosure) with a second substrate layer. For example, FIG. **6**C depicts a cross-section of another example force transducer coupled to a locally-flexible region. The locally-flexible region **600***c* is defined by a substrate layer **614** that is strengthened by a backing plate **616**. In some embodiments, the backing plate can be referred to as a “frame” that provides mechanical support to the substrate layer **614**. The backing plate **616** increases the thickness of certain portions of the substrate layer **614**. In this manner, a relieved section is formed. The relieved section is coupled to a piezoelectric element **606**, as with previously-discussed embodiments.

(91) The foregoing embodiments depicted in FIGS. **6**A-**6**C and the various alternatives thereof and variations thereto are presented, generally, for purposes of explanation, and to facilitate an understanding of various constructions and distributions of force transducers (and, correspondingly, force input sensors and haptic output elements) and the various components thereof, such as described herein. However, it will be apparent to one skilled in the art that some of the specific details presented herein may not be required in order to practice a particular described embodiment, or an equivalent thereof.
(92) Generally and broadly, FIGS. **7-8** depict various example constructions of a force input/haptic output interface. Each depicts a different example arrangement of force transducers relative to a user interface surface.
(93) For example, FIG. **7** depicts an example distribution of force transducers that can be associated with a force input/haptic output interface incorporated into an electronic device **700**. In particular, the electronic device **700** includes an enclosure **702** that defines an external surface **704**. In this embodiment, as with the embodiment depicted in FIG. **2**, the external surface **704** defines an opening to accommodate a user interface substrate **706**. In some examples, the user interface substrate **706** and the external surface **704** may be flush, but this may not be required of all embodiments. For example, in some implementations, the user interface substrate **706** protrudes from the external surface **704**.
(94) In some embodiments, the user interface substrate **706** can be coupled to a frame (not shown) that is disposed within the enclosure **702**. The frame can be made from any number of suitable materials including metals, plastics, glasses, and so on. In one example, the frame is coupled (e.g., via fasteners, adhesive, and so on) to an internal surface of the enclosure **702**.
(95) In the illustrated embodiment, the frame supports four force transducers **708**, **710**, **712**, and **714**. The four force transducers are arranged in a two-by-two grid.
(96) The four force transducers are typically configured and constructed in the same manner, but this is not required. For example, in some embodiments, the force transducer **708** and the force transducer **710** are configured as a first pair of transducers sharing one or more properties whereas the force transducer **712** and the force transducer **714** are configured as a second pair of transducers sharing one or more properties different than the properties of the first pair. For simplicity of description, the description that follows references the force transducer **708**; it is appreciated that the force transducers **710**, **712**, and **714** may be similarly configured.
(97) The force transducer **708** is formed from a piezoelectric material having a crystalline structure that mechanically distorts when an electric field is applied to it. Suitable materials can include lead zirconate titante and potassium sodium niobate. In other examples, the force transducer **708** can be formed from an electroactive polymer, an electromagnetic coil and a ferromagnetic or magnetic plate, or an any combination thereof.
(98) As a result of this construction, compression or expansion of the force transducer **708** induces a deformation in the user interface substrate **706**. Similarly, a compression of the suspended of the user interface substrate **706** induces a compression of the force transducer **708**.
(99) In other embodiments, one or more force transducers of a force input/haptic output interface can be distributed in a different manner. For example, FIG. **8** depicts another sample force input/haptic output interface incorporated into an electronic device **800**. In particular, the electronic device **800** includes an enclosure **802** that includes two rows of force transducers (or haptic actuators or force input sensors), identified as the upper row **804** and the lower row **806**. The upper row **804** and the lower row **806** are coupled to an interior surface of the enclosure, generally below a keyboard. The two rows, as illustrated, each include four force transducers.
(100) The foregoing embodiments depicted in FIGS. **7-8** and the various alternatives thereof and variations thereto are presented, generally, for purposes of explanation, and to facilitate an understanding of various alternative configurations of a force and the various components thereof, such as described herein. However, it will be apparent to one skilled in the art that some of the

specific details presented herein may not be required in order to practice a particular described embodiment, or an equivalent thereof.
(101) For example, it may be understood that, generally and broadly, embodiments described herein can arrange any suitable number of force transducers in any number of suitable patterns including, but not limited to: grid patterns; column-and-row patterns; alternating patterns; repeating patterns; tessellated patterns; offset patterns, and so on.
(102) Further, it may be understood that although the foregoing embodiments reference force transducers implemented with piezoelectric elements, this configuration is not required; in other embodiments, other electroactive elements, materials, or constructions may be used. Suitable materials and constructions may include, without limitation: electroactive polymers; shape-change or memory-wire (e.g., nitinol); ferroelectric polymers; microelectromechanical systems; magnetic attractors; linear actuators; solenoid-based systems; and so on.
(103) Further, as with other embodiments described herein, the force transducers depicted and described in reference to FIGS. **7-8** may not share a uniform thickness with adjacent portions of the frame or the interior surface of an electronic device enclosure. For example, a locally-flexible region can be formed by stiffening or supporting one substrate layer with a second substrate layer. For example, FIG. **9** depicts a cross-section taken through line B-B of FIG. **7**, showing another example force transducer coupled to a locally-flexible region. The locally-flexible region **900** is defined by a substrate layer **902** that can couple to a force transducer **904**. The substrate layer **902** is strengthened by a backing plate **906**. The backing plate **906** increases the thickness of certain portions of the substrate layer **902**, thereby forming a relieved section **908**.
(104) In further examples, a force input/haptic output interface can be integrated within an electronic device in another manner. For example, as noted above, a force input/haptic output interface may be implemented in conjunction with a touch input sensor. FIGS. **10-14** depict examples of such configurations.
(105) For example, FIG. **10** depicts a cross-section (e.g., taken through line B-B of FIG. **7**, showing a different embodiment than that of FIG. **9**) of an example force input/haptic output interface. In this example, the force input/haptic output interface **1000** includes a force input sensor and a haptic output element.
(106) The force input/haptic output interface **1000** is associated with an external cover **1002** that defines an input surface to receive user input (e.g., force and touch) and to provide haptic output. The external cover **1002** is positioned over and coupled to a frame **1004** via a compressible seal **1006**. The compressible seal **1006** can provide relief to the external cover **1002** when a user applies a force to the external cover **1002**. In other cases, the compressible seal **1006** provides an environmental or hermetic seal to protect one or more components internal to the force input/haptic output interface **1000**. In the illustrated example, a touch input sensor **1008** is disposed on an interior surface of the external cover **1002**.
(107) The force input/haptic output interface **1000** also includes a haptic actuator **1010** that is coupled to a reduced-thickness section of the frame **1004**, identified as the reduced-thickness section **1004***a*. More specifically, the haptic actuator **1010** is coupled to a lower surface of the reduced-thickness section **1004***a*. As with other embodiments described herein, the haptic actuator **1010** can include a piezoelectric element.
(108) The frame **1004** can be configured to provide mechanical support to one or more portions of the force input/haptic output interface **1000**. In other examples, the frame **1004** can provide a means of coupling the external cover **1002** to functional portions of the force input/haptic output interface **1000**. More specifically, the frame **1004** can include one or more surfaces configured to adhere to one or more surfaces of the external cover **1002**.
(109) The material or construction of the frame **1004** may be selected at least in part to provide a particular haptic output in response to a compression or deformation of the haptic actuator **1010**. For example, a thickness of the frame **1004** can influence one or more characteristics of a haptic

output generated by the haptic actuator **1010** that is coupled to that frame; a thicker frame may result in attenuation of low frequency outputs from the haptic actuator **1010** whereas a thinner frame may result in attenuation of high frequency outputs from the haptic actuator **1010**.

(110) Similarly, the material of the frame **1004** can influence one or more characteristics of a haptic output generated by the haptic actuator **1010** that is coupled to that frame. For example, an aluminum frame may result in a different haptic output than a copper frame, a plastic frame, or a glass frame.

(111) In many embodiments, the frame **1004** is formed from a different material than the external cover **1002**. For example, the frame **1004** can be formed from metal and the external cover **1002** can be formed from glass.

(112) The force input/haptic output interface **1000** also includes a capacitive force sensor **1012**. The capacitive force sensor **1012** can be defined by two or more electrical contacts separated by a flexible material such as, but not limited to: silicone; plastic; glass; gel; and so on. In some cases, the flexible material of the capacitive force sensor **1012** may have a non-Newtonian response such that when the frame **1004** deforms in response to an actuation of the haptic actuator **1010**, the deformation of the frame **1004** is rigidly translated to the external cover **1002**.

(113) The flexible material of the capacitive force sensor **1012** need not be continuous; in the illustrated example, the flexible material is implemented as three separate flexible dot elements, but this is not required. As a result of this construction, when a force is applied (e.g., by a user) to the external cover **1002**, the flexible material of the capacitive force sensor **1012** can compress, reducing the distance between the two or more electrical contacts and changing the capacitance of the capacitive force sensor **1012**. This change in capacitance can be measured by a controller (that can include drive circuitry, sense circuitry, and the like) coupled to or otherwise in electrical communication with the capacitive force sensor **1012**, which in turn, can correlate the capacitance change (or absolute capacitance) to a magnitude of force applied by the user to the external cover **1002**. In still further examples, the capacitive force sensor **1012** can provide mechanical relief to the external cover **1002**.

(114) In this example, the haptic actuator **1010** can be configured to provide haptic output to a user and the force sensor **1012** can be configured to receive force input from the same user.

(115) In another embodiment, the force input/haptic output interface can be implemented in a different manner. For example, FIG. **11** depicts a cross-section of a force input/haptic output interface **1100**, such as described herein. In this embodiment, the force input/haptic output interface **1000** includes a touch input sensor, a force input sensor, and a haptic output element. In this example, the haptic output element is positioned below the force input sensor.

(116) As with other embodiments described herein, the force input/haptic output interface includes an external cover **1102** that defines an input surface to receive user input and to provide haptic output. The external cover **1102** is positioned over and coupled to a frame **1104** via a compressible seal **1106**, that may be configured and function similarly to compressible seals described in reference to other embodiments herein. A touch input sensor **1108** is disposed on an interior surface of the external cover **1102**. The touch input sensor **1108** can be implemented as a capacitive touch sensor.

(117) As with other embodiments described herein, the force input/haptic output interface **1100** also includes a capacitive force sensor **1110**. The capacitive force sensor **1110** can be defined by two electrical contacts separated by a flexible material such as, but not limited to: silicone, plastic, glass, gel, and so on. The flexible material need not be continuous; in the illustrated example, the flexible material is implemented as a flexible contiguous layer, but this is not required.

(118) The force input/haptic output interface **1100** also includes a stiffener **1112** below the capacitive force sensor **1110**. The stiffener **1112** can be configured and can function similar to stiffeners and backing plates described herein. In particular, the stiffener **1112** may define locally-flexible and/or locally-stiffened regions of the force input/haptic output interface **1100**.

(119) A haptic output element **1114** is positioned below the stiffener **1112**. The haptic output element **1114** is separated from the frame **1104** by a gap **1116**. The gap **1116** is configured to permit the force input/haptic output interface **1100** to flex in response to a force applied by a user or a haptic output generated by the haptic output element **1114**. In other examples, the gap **1116** can be larger or smaller. The gap **1116** may have a uniform or non-uniform thickness. In some cases, the gap **1116** can be filled with a filler material such as, but not limited to: compressible foam; compressible adhesive; compressible liquid; and so on. In other cases, the gap **1116** may be filled with a gas such as air.
(120) FIG. **12** depicts a cross-section of another example force transducer coupled to a substrate associated with a force input/haptic output interface **1200**, such as described herein. The force input/haptic output interface includes an external cover **1202** that defines an input surface to receive user input and to provide haptic output. The external cover **1202** is positioned over and coupled to a frame **1204** via a compressible seal **1206**. As with other embodiments described herein, a touch input sensor **1208** is disposed below an interior surface of the external cover **1202**. In this example, a haptic actuator (described in greater detail below) is coupled below the frame **1204**.
(121) In the illustrated embodiment, a force input sensor **1210** is disposed below the touch input sensor **1208** of the external cover **1202**, such that the force input sensor **1210** experiences strain in proportion to a magnitude of force applied to the external cover **1202**.
(122) The force input sensor **1210** can be constructed in any number of suitable ways. For example, in one embodiment, the force input sensor **1210** is implemented as a piezoelectric sheet configured to compress or expand—and generate a charge measureable as a voltage spike—in response to a force applied to the external cover **1202**. In another embodiment, the force input sensor **1210** is implemented as a strain sensor. A strain sensor be formed from one or more traces of peizoresistive material. In another example, a strain sensor can be an inductive strain sensor configured to exhibit a change in inductance proportional to or otherwise related to a strain experienced by the external cover **1202** in response to a force applied to the external cover **1202**.
(123) The force input sensor **1210** is offset from the frame **1204**, and thereby able to flex, by a separator **1212** (e.g., a spacer). The separator **1212** can be formed from any number of suitable elastic or otherwise flexible materials such as, but not limited to: silicone; plastic; glass; gel; a pressure-sensitive adhesive; and so on. In other cases, the separator **1212** may not be flexible and can be formed form a material such as metal or rigid plastic. In these examples, local or global flexibility of the frame **1204** and/or flexibility of the compressible seal **1206** may provide relief for the force input/haptic output interface **1200** in response to a force input from the user.
(124) In this example, the frame includes a reduced-thickness section **1204***a*. As with other embodiments described herein, the reduced-thickness section **1204***a* can at least partially defined a locally-flexible region of the frame **1204**. The force input/haptic output interface **1200** also includes a haptic actuator **1214** that is coupled to a reduced-thickness section **1204***a* of the frame **1204**. As with other embodiments described herein, the haptic actuator **1214** can include a piezoelectric element.
(125) As a result of the constructed depicted in FIG. **12**, high-voltage electronics that may be required to actuate the haptic actuator **1214** may be physically and structurally isolated from low-voltage electronics that may be required to operate the touch input sensor **1208** and/or the force input sensor **1210**. Further, in some examples, the frame **1204** may be formed from metal. In these examples, the frame **1204** may provide electromagnetic shielding between the haptic actuator **1214** and the force input sensor **1210** and/or the touch input sensor **1208**. As a result of these constructions, a total thickness of the force input/haptic output interface **1200** can be reduced.
(126) FIG. **13** depicts a cross-section of another example force transducer coupled to a substrate associated with a force input/haptic output interface **1300**, such as described herein. The force input/haptic output interface **1300** is configured similar to other embodiments (e.g., FIG. **12**)

described herein. In this example, a force input sensor is coupled to a locally-flexible portion of a frame. The locally-flexible portion of the frame concentrates strain in the force input sensor, thereby increasing the sensitivity of that sensor.

(127) In particular, the force input/haptic output interface **1300** includes an external cover **1302** positioned over a frame **1304** and a compressible seal **1306**. A touch input sensor **1308** is disposed below the external cover **1302**, and is separated from a locally-flexible portion **1312** the frame **1304** by a separator **1310**. A haptic actuator **1314** is coupled to a lower surface of the locally-flexible portion **1312**. The force input/haptic output interface **1300** also includes a force input sensor **1316** that is coupled to the locally-flexible portion **1312**. The force input sensor **1316** can be constructed in any number of suitable ways, such as those described in reference to FIG. **12**. In this example, as with FIG. **12**, electronics associated with the haptic actuator **1314** are physically and structurally isolated from those associated with the touch input sensor **1308**.

(128) In yet another embodiment, a force input sensor and a haptic actuator can be laminated in a single stack below an input surface. FIG. **14** depicts a force input/haptic output interface includes an external cover **1402** coupled to the frame **1404** via a compressible seal **1406**. A touch input sensor **1408** is coupled to an interior surface of the external cover **1402**. A force input sensor **1410** is disposed below the touch input sensor **1408** of the external cover **1402**. In this example, the force input sensor **1410** is backed by a stiffener **1412**. In typical embodiments the stiffener **1412** is formed from a rigid material such as metal, but this may not be required. In other examples, the stiffener **1412** is formed from plastic or glass. In some cases, the stiffener **1412** is formed from metal so as to electromagnetically shield the touch input sensor **1408**.

(129) The force input/haptic output interface **1400** also includes a haptic actuator **1414** that is coupled to a lower side of the stiffener **1412**. The haptic actuator **1414** is separated from a reduced-thickness section **1404***a* of the frame **1404** by a gap **1416**. The gap **1416** permits the stack depicted in FIG. **14** to deform toward the frame **1404** in response to a user input. As with other embodiments described herein, the haptic actuator **1414** can include a piezoelectric element.

(130) Many examples provided above include haptic actuators configured to compress or expand in a direction parallel to a user input surface. However, this may not be required. For example, FIG. **15** depicts a force input/haptic output interface **1500** that includes an external cover **1502** coupled to the frame **1504** via a compressible seal **1506**. As with other embodiments, a touch input sensor **1508** is coupled to an interior surface of the external cover **1502.** A force input sensor (not shown) can be disposed below the touch input sensor **1508** of the external cover **1502**, or in any suitable location.

(131) The force input/haptic output interface **1500** also includes a haptic actuator **1510** that is coupled between the frame **1504** and the touch input sensor **1508**. As a result of this construction, expansion or compression of the haptic actuator **1510** perpendicular to the external cover **1502** results in outward or inward expansion of the external cover **1502**. As with other embodiments described herein, the haptic actuator **1510** can be any suitable haptic actuator such as a piezoelectric actuator, an electroactive polymer actuator, a linear actuator, and so on.

(132) The foregoing embodiments depicted in FIGS. **10-15** and the various alternatives thereof and variations thereto are presented, generally, for purposes of explanation, and to facilitate an understanding of various configurations of a force input/haptic output interface and the various components thereof, such as described herein. However, it will be apparent to one skilled in the art that some of the specific details presented herein may not be required in order to practice a particular described embodiment, or an equivalent thereof.

(133) For example, it may be understood that, generally and broadly, embodiments described herein can arrange various components of a force input/haptic output interface in any suitable order. For example, a force input sensor can be positioned: above, below, or adjacent to a touch input sensor; above, below, or adjacent to a haptic output element; above, below, or adjacent to a frame; above, below, or adjacent to a reduced-thickness section of a frame; above, below, or

adjacent to a cover; and so on. Similarly, a haptic output element can be positioned: above, below, or adjacent to a touch input sensor; above, below, or adjacent to a force input sensor; above, below, or adjacent to a frame; above, below, or adjacent to a reduced-thickness section of a frame; above, below, or adjacent to a cover; and so on. Similarly, a force transducer configured to provide both haptic output and configured to receive force input from a user can be positioned: above, below, or adjacent to a touch input sensor; above, below, or adjacent to a frame; above, below, or adjacent to a reduced-thickness section of a frame; above, below, or adjacent to a cover; and so on.
(134) Generally and broadly, FIGS. **16-18** depict flow charts that correspond to various methods that may be associated with a force input/haptic output interface, such as described above.
(135) FIG. **16** is a flow chart depicting example operations of a method of forming a haptic actuator, such as described herein. The method **1600** begins at operation **1602** in which a reduced-thickness section is formed in or through a substrate. The reduced-thickness section can be formed in any manner including, but not limited to: ablation; abrasion; etching; punching; cutting; and so on. In some cases, the reduced-thickness section can be formed and/or introduced into the substrate while forming the substrate (e.g., molding). In some cases, the reduced-thickness section can be formed by adding material to the substrate adjacent to the region.
(136) The thickness of the substrate can be changed to any suitable thickness relative to adjacent portions of the substrate. In some cases, the reduced-thickness section is an opening. In other cases, the reduced-thickness section has a uniform thickness, whereas in others, the thickness of the reduced-thickness section varies. The reduced-thickness section can take any number of suitable shapes.
(137) Next, at operation **1604**, a force transducer is coupled adjacent to the reduced-thickness section. In some cases, the force transducer is coupled directly to the reduced-thickness section. In other cases, the force transducer is coupled directly to the reduced thickness region.
(138) Next, at operation **1606**, the force transducer can be coupled to drive circuitry and/or sense circuitry. In many cases, the drive circuitry and the sense circuitry are implemented as different portions of a single controller in electrical communication or otherwise coupled to the force transducer.
(139) FIG. **17** is a flow chart depicting example operations of another method of forming a haptic actuator, such as described herein. The method **1700** beings at operation **1702** in which an instruction to provide haptic feedback is received. In typical embodiments, the instruction is received by a controller in signal communication with a set of force transducers or haptic actuators.
(140) At operation **1704**, a set or subset of haptic actuators of a set of haptic actuators (or force transducers) is selected. In some cases, a single haptic actuator can be selected. In other cases, more than one haptic actuator can be selected. In still further embodiments, all haptic actuators of a set of haptic actuators can be selected.
(141) At operation **1706**, the set or subset of haptic actuators of the set of haptic actuators selected at operation **1704** can be actuated or driven by a controller or drive circuitry. In some cases, a drive signal corresponding to each individual selected haptic actuator can be applied to the respective haptic actuator. In some cases, the drive signal(s) applied to each haptic actuator are the same, whereas in others the drive signal(s) may be unique to each haptic actuator of the set or subset of haptic actuators of the set of haptic actuators selected at operation **1704**.
(142) In one embodiment, the drive signals are configured to generate haptic outputs that constructively interfere with one another at a particular location or selected location of an input or user interface surface associated with or coupled to the haptic actuators. In other cases, the drives signals are configured to generate haptic outputs that destructively interfere with one another at a particular location or more than one location of the input or user interface surface. In other cases, the drive signals are configured to vibrate a region of the input or user interface surface at a particular frequency.
(143) It may be appreciated that the specific examples listed above are not exhaustive; any number

of suitable drive signals can be applied to any number of haptic actuators.
(144) FIG. **18** is a flow chart depicting example operations of a method of receiving force input. The method **1800** begins at operation **1802** in which an indication that a user is touching an input surface or user interface surface is received. Next, at operation **1804**, output from a force input sensor and/or a force transducer is received. Finally, at operation **1806**, the output can be correlated to a non-binary magnitude of force applied to the input surface or user interface surface.
(145) As noted above, many embodiments described herein reference a force input/haptic output interface operated in conjunction with a non-display region of a portable electronic device, such as a trackpad of a laptop computer. It may be appreciated, however, that this is merely one example; other configurations, implementations, and constructions are contemplated in view of the various principles and methods of operation, and reasonable alternatives thereto, described in reference to the embodiments described above.
(146) For example, without limitation, a force input/haptic output interface can be additionally or alternatively associated with: a display surface; an enclosure or enclosure surface; a planar surface; a curved surface; an electrically conductive surface; an electrically insulating surface; a rigid surface; a flexible surface; a key cap surface; a trackpad surface; a display surface; and so on. The user interface surface can be, without limitation: a front surface; a back surface; a sidewall surface; or any suitable surface of an electronic device or electronic device accessory. Typically, the user interface surface of a force input/haptic output interface is an external surface of the associated portable electronic device but this may not be required.
(147) Further, although many embodiments reference a force input/haptic output interface positioned in a portable electronic device (such as a cell phone or tablet computer) it may be appreciated that a force input/haptic output interface can be incorporated into any suitable electronic device, system, or accessory including but not limited to: portable electronic devices (e.g., battery-powered, wirelessly-powered devices, tethered devices, and so on); stationary electronic devices; control devices (e.g., home automation devices, industrial automation devices, aeronautical or terrestrial vehicle control devices, and so on); personal computing devices (e.g., cellular devices, tablet devices, laptop devices, desktop devices, and so on); wearable devices (e.g., implanted devices, wrist-worn devices, eyeglass devices, and so on); accessory devices (e.g., protective covers such as keyboard covers for tablet computers, stylus input devices, charging devices, and so on); and so on.
(148) One may appreciate that although many embodiments are disclosed above, that the operations and steps presented with respect to methods and techniques described herein are meant as exemplary and accordingly are not exhaustive. One may further appreciate that alternate step order or, fewer or additional operations may be required or desired for particular embodiments.
(149) For example, as noted above a force transducer can be configured in a number of suitable ways. An example force transducer includes a ground electrode and a drive and/or sense electrode separated by a body formed from a material that is configured to contract or expand in the presence of an electric field and, additionally, develop a measureable charge in response to compression or strain. Suitable materials include, but may not be limited to: lead-based piezoelectric alloys; non-leaded materials such as metal niobates or barium titanate; in addition to electroactive polymers. In other examples, other piezoelectric compositions or multi-layered or interdigitated elements can be selected.
(150) In many embodiments, a force transducer is communicably coupled to a controller that includes drive circuitry and/or sense circuitry. The drive circuit is configured to apply a drive signal to the drive and/or sense electrode of the force transducer. The drive signal can be any suitable signal including, but not limited to: a voltage bias; a voltage signal; an alternating signal; and so on. In some cases, the drive signal has an arbitrary waveform. The drive circuit can include one or more signal processing stages that may be used to generate, augment, or smooth the drive signal. For example, the drive circuit can include one or more of, without limitation and in no particular

order: amplifying stages; filtering stage; multiplexing stages; digital-to-analog conversion stages; analog-to-digital conversion stages; comparison stages; feedback stages; and so on. The drive circuit can be implemented with analog circuit components, digital circuit components, passive circuit components, and/or active circuit components. In some examples, the drive circuit is implemented as an integrated circuit.

(151) As with the drive circuit, the sense circuit, configured to receive voltage signals from the force transducer, can be implemented in any number of suitable ways. In many examples, the sense circuit includes one or more signal processing stages that may be used to receive, amplify, augment, or smooth a sense signal obtained from one or more of the sense electrodes. For example, the sense circuit can include one or more of, without limitation and in no particular order: amplifying stages; filtering stages; multiplexing stages; digital-to-analog conversion stages; analog-to-digital conversion stages; comparison stages; feedback stages; and so on. The sense circuit can be implemented with analog circuit components, digital circuit components, passive circuit components, and/or active circuit components. In some examples, the sense circuit is implemented as a single integrated circuit.

(152) Once a sense signal (or more than one sense signal) is obtained, the sense circuit can correlate the one or more properties of the sense signal(s) to an amount of compression or strain experienced by the force transducer. In other cases, the sense circuit can correlate the measurement directly to an amount of force applied to the force transducer.

(153) In many cases, drive circuitry and sense circuitry (or, more generally, a “controller”) can be configured to operate a haptic actuator and a force input sensor simultaneously. In these examples, sense circuitry can be configured to adjust an output received from a force input sensor bases on an operation of drive circuitry. For example, in one embodiment, a sense signal received by sense circuitry may have an increased amplitude due to a simultaneously or substantially simultaneous actuation of a haptic actuator. In this example, the sense circuitry can be configured to adjust and/or filter the sense signal based on the drive signal. In other cases, a sense signal received by sense circuitry may have a decreased amplitude due to a simultaneously or substantially simultaneous actuation of a haptic actuator. In this example, the sense circuitry can compensate for effects of the actuation by increasing the amplitude of the sense signal based on the drive signal. In still other cases, the sense signal and the drive signal may be out-of-band with respect to one another. For example, a drive signal may have a central frequency between 100 Hz and 250 Hz whereas a sense signal may have a central frequency between 0 Hz and 10 Hz. In these examples, the sense circuitry may be configured with a low-pass filter.

(154) It may be appreciated that any implementation of any embodiment described herein can be configured in this manner. For example, with reference to the electronic device **100** depicted in FIG. **1**, the set of force transducers of the force input/haptic output interface **110** can be used to user provide haptic output to a user and/or to receive force input from the user. For example, in a first mode (e.g., a “drive mode” or a “haptic mode”), the set of force transducers can be operated individually or collectively (e.g., constructively or destructively interfering at one or more locations) to produce a haptic output through the user interface surface. The haptic output can be any suitable output, such as a click, vibration, lateral shift, vertical shift, and so on.

(155) In many cases, a haptic output is generated in conjunction with a function or operation of the electronic device **100**. For example, a user of the electronic device **100** can receive a first haptic output after selecting (e.g., with a cursor) a first element rendered in a graphical user interface displayed by the primary display **104**. The user can receive a second haptic output after selecting a second element rendered in the same graphical user interface. The first or second haptic outputs may be transient or sustained.

(156) In some cases, one or more characteristics of a haptic output generated by the set of force transducers can depend upon a mode, state, application, function, setting, operation, or task of the electronic device **100**. A haptic characteristic can include, but may not be limited to: duration;

intensity; amplitude; sound; location; type of displacement; frequency of vibration; amount of ring-down compensation; and so on. Any suitable haptic characteristic can be used.

(157) In further cases, more than one haptic output can be generated by the set of force transducers at a particular time. For example, a first force transducer located at a first location of the user interface surface can generate a first haptic output while a second force transducer located at a second location of the user interface surface can generate a second haptic output. The different haptic outputs may be associated with different locations of the user interface surface.

(158) In another example, in a second mode (e.g., a “sense mode” or an “input mode”), the set of force transducers can be operated individually or collectively to produce an electrical signal in response to a force input applied to the user interface surface. The electrical signal can be correlated (e.g., by a controller or circuitry in signal communication with the set of force transducers) to a magnitude and/or location of force applied by one or more objects in contact with the user interface surface. In some examples, the force input/haptic output interface **110** can be operated in the sense mode and the drive mode simultaneously. In other cases, the force input/haptic output interface **110** can alternatively operate in the drive mode and the sense mode.

(159) Further to the foregoing, although the disclosure above is described in terms of various exemplary embodiments and implementations, it should be understood that the various features, aspects and functionality described in one or more of the individual embodiments are not limited in their applicability to the particular embodiment with which they are described, but instead can be applied, alone or in various combinations, to one or more of the some embodiments of the invention, whether or not such embodiments are described and whether or not such features are presented as being a part of a described embodiment. Thus, the breadth and scope of the present invention should not be limited by any of the above-described exemplary embodiments but is instead defined by the claims herein presented.

Claims

1. An electronic device comprising: an enclosure having an external surface comprising: a first region in which a keyboard is positioned, the keyboard having a length; and a second region positioned adjacent to the keyboard and extending across at least the length of the keyboard, the second region being continuous and substantially planar, and the second region comprising a first set of locally-flexible areas and a second set of locally-flexible areas defined by openings in an internal surface of the enclosure opposite the external surface, wherein the second region comprises a user input surface coinciding with at least one of the first set of locally-flexible areas or the second set of locally-flexible areas, the user input surface being visually continuous with a surface of the first region; a set of sensors within the second region and extending across at least the length of the keyboard; and a haptic output interface operable to provide haptic feedback at the user input surface via at least one of the first set of locally-flexible areas or the second set of locally-flexible areas in response to user input detected by the set of sensors.
2. The electronic device of claim 1, wherein the first set of locally-flexible areas is more rigid than the second set of locally-flexible areas.
3. The electronic device of claim 1, wherein there are more locally-flexible areas in the second set of locally-flexible areas than in the first set of locally flexible areas.
4. The electronic device of claim 1, wherein the set of sensors is sensitive to a change in capacitance across the second region.
5. The electronic device of claim 1, wherein the set of sensors positioned in the second region correspond to the first and second sets of locally-flexible areas of the second region.
6. The electronic device of claim 1, further comprising a frame defined into or coupled to an interior surface of the enclosure at the second region.
7. The electronic device of claim 1, wherein the second region extends across an entirety of a width

of the external surface.
8. A computing device, comprising: an enclosure; a keyboard positioned in the enclosure; a display positioned in the enclosure on a first side of the keyboard; a user interface surface positioned on the enclosure on a second side of the keyboard, the second side being opposite the first side, the user interface surface extending across an entirety of a width of the enclosure, and the user interface surface including a pair of locally movable regions positioned adjacent to opposing ends of the width of the enclosure and at least one additional locally movable region positioned between the pair of locally movable regions, at least one of the pair of locally movable regions or the at least one additional locally movable region comprising a recess in a material defining the user interface surface, wherein the user interface surface and the enclosure share an unbroken, visually continuous outer surface; and a set of sensors operable to receive user input via the user interface surface across the entirety of the width of the enclosure, at least some sensors of the set of sensors being disposed at the pair of locally movable regions and the at least one additional locally movable region.
9. The computing device of claim 8, wherein the set of sensors is configured to sense a change in capacitance in response to the user input.
10. The computing device of claim 8, further comprising a haptic output device operable to provide haptic output in response to the user input.
11. The computing device of claim 10, wherein the haptic output is provided across the entirety of the width of the enclosure.
12. The computing device of claim 8, wherein the pair of locally movable regions is different from the at least one additional locally movable region positioned between the pair of locally movable regions.
13. The computing device of claim 8, wherein the set of sensors is arranged in multiple rows.
14. The computing device of claim 8, wherein the set of sensors correlate to a set of deformable portions of the enclosure in the user interface surface.
15. A laptop computer, comprising: an upper enclosure; a display positioned in the upper enclosure; a lower enclosure pivotally coupled with the upper enclosure via a hinge; a keyboard positioned in the lower enclosure; a trackpad area in the lower enclosure positioned on a side of the keyboard opposite the display, the trackpad area extending across at least a width of the keyboard parallel to the hinge, the trackpad area comprising a reduced-thickness section of the lower enclosure; and a haptic output device directly coupled to the lower enclosure in the reduced-thickness section, wherein an external surface of the lower enclosure covers the trackpad area, wherein the external surface and trackpad area are visually continuous and uninterrupted across the lower enclosure over and around the trackpad area, the external surface and the trackpad area being coplanar.
16. The laptop computer of claim 15, further comprising locally-flexible regions formed into an interior surface of the lower enclosure below a portion of the external surface corresponding to the trackpad area, the locally-flexible regions comprising the reduced thickness section of the lower enclosure.
17. The laptop computer of claim 15, wherein the external surface of the lower enclosure comprises a glass top surface, wherein the trackpad area is defined in the glass top surface.
18. The laptop computer of claim 15, wherein the trackpad area comprises a capacitive touch sensor operable to detect a change in capacitance at the trackpad area.
19. The laptop computer of claim 15, wherein the trackpad area comprises a set of force sensors operable to detect a force applied to the trackpad area of the lower enclosure.
20. The laptop computer of claim 15, wherein the haptic output device is operable to provide haptic output through the trackpad area across at least the width of the keyboard.